

Total No. of Questions : 8]

SEAT No. :

PB2501

[6263]-387

[Total No. of Pages : 2

B.E. (A.I. & D.S.)

INFORMATION RETRIEVAL

(2019 Pattern) (Semester - VII) (Elective - IV) (417524 B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary

Q1) a) Define and explain Probabilistic Retrieval. [4]

b) Define the following with respect to Information Retrieval: [6]

i) Vector Space Model

ii) Term Frequency

iii) Inverse Document Frequency

c) What is the binary independent retrieval model? [6]

OR

Q2) a) List the problems associated with n-gram. Explain how these problems are handled. [6]

b) List and explain the challenges in Natural language processing. [4]

c) Explain with suitable example different levels of NLP. [6]

Q3) a) What is the difference between clustering and classification? Can clustering be used for classification purposes? [6]

b) What are different types of clustering algorithm? Explain any one of them. [6]

c) Explain in detail about naïve Bayes algorithm and its application in text classification. [6]

OR

Q4) a) Write partitioning algorithm for clustering. [10]

b) Solve Agglomerative hierarchical clustering for single link with example. [8]

P.T.O.

- Q5)** a) Discuss the factors that influence a webpages PageRank score. Discuss the challenges involve in web search engine. [8]
- b) What is page ranking? Calculate page rank of following web pages. Assume damping factor 0.7. [10]



OR

- Q6)** a) Define Python library is used for web crawling? [6]
- b) What are the main challenges posed by Web? [6]
- c) Explain the components of focused web crawlers? [6]

- Q7)** a) Outline the steps to build a recommender system with an example. [10]
- b) What are the four phases of recommender system? [8]

OR

- Q8)** a) Multimedia Information Retrieval [8]
- b) What are the different evaluation metric for Recommender system. [10]

